PARISHRAM 2.0 2026 (Class 12th)

Maths

DPP: 4

Matrices

Q1 The matrix $\begin{bmatrix} 0 & -5 & 8 \\ 5 & 0 & 12 \\ -8 & -12 & 0 \end{bmatrix}$ is a

(A) symmetric matrix

(B) scalar matrix

(C) diagonal matrix

(D) skew-symmetric matrix

Q2 Which option represents the matrix $\mathbf{B} = \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix}$ as the sum of a symmetric and a skew symmetric matrix.

(A) $\begin{bmatrix} 2 & -1 & -1 \\ -5 & 2 & 1 \\ -\frac{3}{2} & 1 & -7 \end{bmatrix} + \begin{bmatrix} 0 & -\frac{1}{2} & 7 \\ 2 & 0 & 3 \\ 5 & -3 & 0 \end{bmatrix}$

(B) $\begin{bmatrix} 2 & -\frac{3}{2} & -\frac{3}{2} \\ -\frac{3}{2} & 3 & 1 \\ -\frac{3}{2} & 1 & -3 \end{bmatrix} + \begin{bmatrix} 0 & -\frac{1}{2} & -\frac{5}{2} \\ \frac{1}{2} & 0 & 3 \\ \frac{5}{2} & -3 & 0 \end{bmatrix}$

(C) $\begin{bmatrix} 2 & -\frac{3}{2} & 7 \\ -2 & 3 & 5 \\ -2 & 7 & -3 \end{bmatrix} + \begin{bmatrix} 0 & -1 & -5 \\ \frac{1}{2} & 1 & 3 \\ 5 & -1 & 1 \end{bmatrix}$

(D) $\begin{bmatrix} 5 & -3 & -3 \\ 6 & 3 & 2 \\ -1 & 1 & -4 \end{bmatrix} + \begin{bmatrix} 1 & -1 & -3 \\ 2 & 1 & 3 \\ 2 & -3 & 1 \end{bmatrix}$

Q3 If the matrix $A = \begin{bmatrix} 0 & a & 3 \\ 2 & b & -1 \\ c & 1 & 0 \end{bmatrix}$ is a skew-symmetric matrix, then $a + b + c =$

(A) 5 (B) 0

(C) –5 (D) None of these

Q4 $\mathbf{A} = \begin{bmatrix} 3 & \mathbf{a} & -1 \\ 2 & 5 & \mathbf{c} \\ \mathbf{b} & 8 & 2 \end{bmatrix}$ is symmetric and $\mathbf{B} = \begin{bmatrix} \mathbf{d} & 3 & \mathbf{a} \\ \mathbf{b-a} & \mathbf{e} & \mathbf{-2b-c} \\ -2 & 6 & -\mathbf{f} \end{bmatrix}$ is skew-symmetric, then the value of AB is:

(A) $\begin{bmatrix} -4 & 3 & -6 \\ -31 & 54 & -26 \\ -28 & 9 & 50 \end{bmatrix}$

(B) $\begin{bmatrix} -4 & 3 & -6 \\ 31 & 54 & -26 \\ -28 & 9 & 50 \end{bmatrix}$

(C) $\begin{bmatrix} -4 & 3 & -6 \\ -31 & 54 & -26 \\ -28 & 9 & -50 \end{bmatrix}$

(D) None of these

Q5 If A, B are symmetric matrices of same order, than AB – BA is a

(A) Skew symmetric matrix

(B) Symmetric matrix

(C) Zero matrix

(D) Identity matrix

Q6 If A and B are square matrices of the same order, then matrix $\left(AB^T - BA^T\right)$ is a

(A) symmetric matrix

(B) skew-symmetric matrix

(C) null matrix

(D) unit matrix

Q7 Assertion: If A is a symmetric matrix, then B'AB is also symmetric.

Reason: (ABC)' = C'B'A'

(A) Both assertion and reason are correct and reason is the correct explanation for assertion.
(B) Both assertion and reason are correct but Reason is not the correct explanation for assertion.
(C) Assertion is correct but reason is incorrect.
(D) Assertion is incorrect and reason is correct.

Q8 Assertion: If A is a symmetric matrix, then A^3 is a skew-symmetric matrix.
Reason: If A is a skew-symmetric matrix, then A^2 is a symmetric matrix.
(A) Assertion is true, Reason is true; Reason is a correct explanation for Assertion.
(B) Assertion is true, Reason is true; Reason is not a correct explanation for Assertion.
(C) Assertion is true; Reason is false.
(D) Assertion is false; Reason is true.

Answer Key

Q1	D
Q2	B
Q3	C
Q4	C
Q5	A
Q6	B
Q7	A
Q8	D

Hints & Solutions

Note: scan the QR code to watch video solution

Q1 Text Solution:

We have $\mathrm{A}=\begin{bmatrix}0 & -5 & 8\\ 5 & 0 & 12\\ -8 & -12 & 0\end{bmatrix}$

$\therefore \mathrm{A'}=\begin{bmatrix}0 & 5 & -8\\ -5 & 0 & -12\\ 8 & 12 & 0\end{bmatrix}=-\mathrm{A}$

So, matrix A is skew-symmetric.

Video Solution:

Q2 Text Solution:

Here

$\mathrm{B'}=\begin{bmatrix}2 & -1 & 1\\ -2 & 3 & -2\\ -4 & 4 & -3\end{bmatrix}$

Let

$\mathrm{P}=\frac{1}{2}\left(\mathrm{B}+\mathrm{B'}\right)=\frac{1}{2}\begin{bmatrix}4 & -3 & -3\\ -3 & 6 & 2\\ -3 & 2 & -6\end{bmatrix}$

$=\begin{bmatrix}2 & \frac{-3}{2} & \frac{-3}{2}\\ \frac{-3}{2} & 3 & 1\\ \frac{-3}{2} & 1 & -3\end{bmatrix}$

Now $\mathrm{P'}=\begin{bmatrix}2 & \frac{-3}{2} & \frac{-3}{2}\\ \frac{-3}{2} & 3 & 1\\ \frac{-3}{2} & 1 & -3\end{bmatrix}=\mathrm{P}$

Thus $\mathrm{P}=\frac{1}{2}\left(\mathrm{B}+\mathrm{B'}\right)$ is a symmetric matrix.

Also, let

$\mathrm{Q}=\frac{1}{2}\left(\mathrm{B}-\mathrm{B'}\right)=\frac{1}{2}\begin{bmatrix}0 & -1 & -5\\ 1 & 0 & 6\\ 5 & -6 & 0\end{bmatrix}$

$=\begin{bmatrix}0 & \frac{-1}{2} & \frac{-5}{2}\\ \frac{1}{2} & 0 & 3\\ \frac{5}{2} & -3 & 0\end{bmatrix}$

Then $\mathrm{Q'}=\begin{bmatrix}0 & \frac{1}{2} & \frac{5}{3}\\ \frac{-1}{2} & 0 & -3\\ \frac{-5}{2} & 3 & 0\end{bmatrix}=-\mathrm{Q}$

Thus $\mathrm{Q}=\frac{1}{2}\left(\mathrm{B}-\mathrm{B'}\right)$ is a skew symmetric matrix.

Now

$\mathrm{P}+\mathrm{Q}=\begin{bmatrix}2 & \frac{-3}{2} & \frac{-3}{2}\\ \frac{-3}{2} & 3 & 1\\ \frac{-3}{2} & 1 & -3\end{bmatrix}$

$+\begin{bmatrix}0 & \frac{-1}{2} & \frac{-5}{2}\\ \frac{1}{2} & 0 & 3\\ \frac{5}{2} & -3 & 0\end{bmatrix}=\begin{bmatrix}2 & -2 & -4\\ -1 & 3 & 4\\ 1 & -2 & -3\end{bmatrix}=\mathrm{B}$

Thus, B is represented as the sum of a symmetric and a skew symmetric matrix.

Video Solution:

Q3 Text Solution:

A is a skew-symmetric matrix.

$\therefore \quad a=-2,\ b=0 \text{ and } c=-3 \Rightarrow a+b+c=-5$

Video Solution:

Q4 Text Solution:

A is symmetric

$$\Rightarrow A^T = A$$

$$\Rightarrow \begin{bmatrix} 3 & 2 & b \\ a & 5 & 8 \\ -1 & c & 2 \end{bmatrix} = \begin{bmatrix} 3 & a & -1 \\ 2 & 5 & c \\ b & 8 & 2 \end{bmatrix}$$

$$\Rightarrow a = 2,\ b = -1,\ c = 8$$

B is skew-symmetric

$$\Rightarrow B^T = -B$$

$$\Rightarrow \begin{bmatrix} d & b-a & -2 \\ 3 & e & 6 \\ a & -2b-c & -f \end{bmatrix}$$

$$= \begin{bmatrix} -d & -3 & -a \\ a-b & -e & 2b+c \\ 2 & -6 & f \end{bmatrix}$$

$$\Rightarrow d = -d,\ f = -f \text{ and } e = -e$$

$$\Rightarrow d = e = f = 0$$

$$\text{So } A = \begin{bmatrix} 3 & 2 & -1 \\ 2 & 5 & 8 \\ -1 & 8 & 2 \end{bmatrix} \text{ and } B$$

$$= \begin{bmatrix} 0 & 3 & 2 \\ -3 & 0 & -6 \\ -2 & 6 & 0 \end{bmatrix}$$

$$\Rightarrow AB = \begin{bmatrix} 3 & 2 & -1 \\ 2 & 5 & 8 \\ -1 & 8 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 3 & 2 \\ -3 & 0 & -6 \\ -2 & 6 & 0 \end{bmatrix} =$$

$$\begin{bmatrix} -4 & 3 & -6 \\ -31 & 54 & -26 \\ -28 & 9 & -50 \end{bmatrix}$$

Video Solution:

Q5 Text Solution:

A and B are symmetric matrrices, therefore, we have:

$$A' = A \text{ and } B' = B \quad ...(1)$$

Consider $(AB-BA)' = (AB)' - (BA)'$

$[(A-B)' = A'-B']$

$$= B'A' - A'B' \qquad [(AB)' = B'A']$$

$$= BA - AB \qquad [\text{by } (1)]$$

$$= -(AB - BA)$$

$$\therefore (AB - BA)' = -(AB - BA)$$

Thus, (AB – BA) is a skew-symmetric matrix.

Video Solution:

Q6 Text Solution:

As

$$(AB^T - BA^T)^T = (AB^T)^T - (BA^T)^T$$

$$= (B^T)^T A^T - (A^T)^T B^T$$

$$= BA^T - AB^T = -(AB^T - BA^T)$$

$\therefore\ AB^T - BA^T$ is a skew-symmetric matrix.

Video Solution:

Q7 Text Solution:

For three matrices A, B an C, if ABC is define then (ABC)' = C'B'A'.

Hence, Reason (R) is correct

Given that A is symmetric A' = A

$$(\mathrm{B'AB})' = \mathrm{B'A'(B')'} = \mathrm{B'AB}.$$

Therefore, $\mathrm{B'AB}$ is a symmetric matrix.

Hence, Assertion (A) is correct.

Video Solution:

Q8 **Text Solution:**

All positive integral powers of a symmetric matrix are symmetric matrices. So, Assertion is not true. All positive even integral powers of a skew-symmetric matrix are symmetric matrices. So, Reason is true. Hence, Assertion is false; Reason is true.

Video Solution:

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